

Abhinav Naidu Chintala

📍 Hyderabad, India ✉ abhinavch017@gmail.com ☎ +91 8978687566 🌐 abhinav-portfolio.com
in abhinavchintala 🔑 abhinav00017

Education

GITAM University

Computer Science Engineering

June 2019 – April 2023

- GPA: 9.0/10

Experience

Software Engineer

Tata Consultancy Services - TCS

Hyderabad, TS

June 2023 – Present

- Co-led in designing and development of an automated, full-stack cloud resource management system using Python, Flask, Docker, and Azure APIs, reducing deployment time by **80%** and improving resource deployment efficiency.
- Developed RESTful APIs as microservice using FastAPI, implementing effective API design, security, and documentation with Swagger/OpenAPI, resulting in **30% faster deployments**.
- Engineered end-to-end FinOps automation pipelines utilizing **AWS Lambda, S3, and Athena** for ETL, enabling data-driven decisions and cost optimization across AWS and Azure clouds, and **reducing cloud spend by 15%**.
- Enabled **\$2M+** in cloud cost savings by automating VM right-sizing and storage optimization across AWS and Azure.
- Built an end-to-end **certificate generation tool** for an insurance client using Flask, NoSQL, and Svelte, deployed on **Zoho Catalyst**, featuring employee login for policy selection and customer **OTP-based login** to securely fill certificate holder details, reducing manual errors and improving operational efficiency.
- Collaborated in building an end-to-end application with **three role-based logins** (admin with dashboards, producer, and customer) using **React and Python Flask**, enhancing accessibility and operational workflows across different user groups.
- Designed and implemented a proof of concept (POC) for an **AI-powered CI/CD assistant** by integrating OpenAI APIs with Azure DevOps, automating containerized application deployments using Docker to enhance deployment speed.
- Led the design and execution of a scalable data migration, **centralizing Azure network IP** data using secure RESTful APIs, improving network data accessibility, reliability, and operational efficiency.
- Added a change request feature for processed resource workflows, improving **user experience and usability** through thoughtful UI/UX design and backend logic enhancements.
- Applied Agile software development practices and Git-based version control to collaborate with cross-functional teams in delivering production-grade software, ensuring high unit test coverage.

Projects

VisaPro V1 & V2

github.com/visapro.git 

- Designed and developed a full-stack **visa processing platform** enabling international travelers to retrieve dynamic visa requirements and assess eligibility based on country, age, and travel purpose.
- Built a modular, scalable architecture using **FastAPI (Python)** for backend services and **React.js with state management** for a responsive, interactive frontend, enabling extensibility for multi-language support and third-party integrations.
- Integrated an **AI-driven virtual assistant** powered by OpenAI's GPT APIs to deliver **real-time, context-aware responses**, improving user engagement and reducing manual information search time.
- Developed features for **user authentication, secure session management**, and CRUD operations using SQLite, ensuring data security and consistent user experience.

- **Tech Stack:** React.js, HTML, CSS, FastAPI, REST APIs, SQLite, OpenAI API, JWT Authentication.

Computer Vision-based Crime Detection System (Published Research)

- Built a deep learning-based **video surveillance system** to detect criminal activities in real time using a **CNN-LSTM architecture**, with frame-wise activity classification and alert triggers.
- Trained the model on the **UCF-50 dataset**, applying data augmentation and preprocessing to improve model generalization for violent activity detection.
- Designed a **real-time inference pipeline** using OpenCV and Keras/TensorFlow to automatically flag suspicious behavior and notify authorities through alert mechanisms.
- Automated activity recognition to assist human operators, increasing surveillance efficiency and reducing false negatives in crime detection.

Skills

- Programming Languages: *Advanced Python, SQL.*
- Web Frameworks: *Flask, FastAPI's, Django, and RESTful API.*
- Front-End Technologies: *JavaScript, HTML5, CSS3, React.js, and basic Node.js.*
- Databases: *MySQL, PostgreSQL, MongoDB.*
- Cloud Platforms: Experience with *Microsoft Azure, AWS.*
- Development Tools & Platforms: *Git, Azure DevOps, Jira, Docker, Kubernetes, CI/CD Pipelines.*

Courses

- Machine Learning & Data Science: With *regression, SVM, KNN, Naïve Bayes, Decision Tree, Random Forest, Gradient Boost, Deep Learning, Neural Networks (CNN, RNN), and Natural Language Processing (NLP).*
- Software Development Principles: Expertise in *Object-Relational Mapping (ORM), software security, performance optimization, and scalable application design.*

Certifications

- Microsoft Certified: AI-900 Azure AI Fundamentals – *2024*
- Microsoft Certified: DP-900 Azure Data Fundamentals – *2023*
- Microsoft Certified: AZ-900 Azure Cloud Fundamentals – *2022*
- GenAI E0 Certification by Google - *Sep-2023*